

Modul Praktikum 05

Sistem Informasi Geografis (SIG)

Geo-Databases: Studi Kasus Kawasan UMKT Samarinda

14 April 2026

1 Tujuan Instruksional

Setelah menyelesaikan modul ini, mahasiswa diharapkan mampu:

- Melakukan konfigurasi Cloud Database (Supabase/Neon) dengan ekstensi PostGIS.
- Mengelola tipe data spasial berbasis koordinat lokal Samarinda.
- Melakukan analisis *Proximity* dan *Geofencing* di sekitar Kampus UMKT.

2 Persiapan Perangkat

Gunakan layanan **Supabase** untuk praktikum ini. Aktifkan ekstensi spasial pada SQL Editor Anda:

```
1 -- Mengaktifkan 'otak' spasial pada database PostgreSQL
2 CREATE EXTENSION postgis;
```

3 Latihan 1: DDL Spasial Kawasan UMKT

Kita akan merepresentasikan infrastruktur kampus dan sekitarnya dalam database.

3.1 Tabel Point: Fasilitas Kampus UMKT

```
1 CREATE TABLE fasilitas_umkt (
2     id SERIAL PRIMARY KEY,
3     nama_tempat VARCHAR(100),
4     kategori VARCHAR(50),
5     geom GEOMETRY(Point, 4326)
6 );
7
8 -- Memasukkan data koordinat Kampus UMKT (Lon, Lat)
```

```

9 INSERT INTO fasilitas_umkt (nama_tempat, kategori, geom) VALUES
10 ('Gedung E UMKT', 'Pendidikan', ST_GeomFromText('POINT(117.15341
    -0.48512)', 4326)),
11 ('Masjid Al-Ikhlash UMKT', 'Ibadah', ST_GeomFromText('POINT
    (117.15375 -0.48545)', 4326)),
12 ('Gedung Puskom', 'Fasilitas', ST_GeomFromText('POINT(117.15290
    -0.48490)', 4326));

```

3.2 Tabel LineString: Akses Jalan Ir. H. Juanda

```

1 CREATE TABLE jalan_samarinda (
2     id SERIAL PRIMARY KEY,
3     nama_jalan VARCHAR(100),
4     geom GEOMETRY(LineString, 4326)
5 );
6
7 INSERT INTO jalan_samarinda (nama_jalan, geom) VALUES
8 ('Jl. Ir. H. Juanda', ST_GeomFromText('LINESTRING(117.1520 -0.4860,
    117.1550 -0.4845)', 4326));

```

3.3 Tabel Polygon: Plot Area Kampus UMKT

```

1 CREATE TABLE batas_wilayah (
2     id SERIAL PRIMARY KEY,
3     nama_area VARCHAR(50),
4     geom GEOMETRY(Polygon, 4326)
5 );
6
7 INSERT INTO batas_wilayah (nama_area, geom) VALUES
8 ('Kawasan Utama UMKT', ST_GeomFromText('POLYGON((117.1525 -0.4840,
    117.1545 -0.4840, 117.1545 -0.4865, 117.1525 -0.4865, 117.1525
    -0.4840))', 4326));

```

4 Data

4.1 Dataset Lengkap: Fasilitas UMKT (25 Points)

```

1 INSERT INTO fasilitas_umkt (nama_tempat, kategori, geom) VALUES
2 ('Gedung A (Rektorat)', 'Pendidikan', ST_GeomFromText('POINT
    (117.1531 -0.4852)', 4326)),
3 ('Gedung B', 'Pendidikan', ST_GeomFromText('POINT(117.1533 -0.4855)
    ', 4326)),
4 ('Gedung C', 'Pendidikan', ST_GeomFromText('POINT(117.1535 -0.4858)
    ', 4326)),
5 ('Gedung D', 'Pendidikan', ST_GeomFromText('POINT(117.1537 -0.4861)
    ', 4326)),
6 ('Gedung E', 'Pendidikan', ST_GeomFromText('POINT(117.1539 -0.4864)
    ', 4326)),

```

```

7 ('Gedung F', 'Pendidikan', ST_GeomFromText('POINT(117.1541 -0.4867)
  ', 4326)),
8 ('Gedung G', 'Pendidikan', ST_GeomFromText('POINT(117.1543 -0.4870)
  ', 4326)),
9 ('Perpustakaan UMKT', 'Edukasi', ST_GeomFromText('POINT(117.1528
  -0.4850)', 4326)),
10 ('Masjid Al-Ikhlas', 'Ibadah', ST_GeomFromText('POINT(117.1537
  -0.4854)', 4326)),
11 ('Laboratorium Komputer', 'Laboratorium', ST_GeomFromText('POINT
  (117.1529 -0.4848)', 4326)),
12 ('Laboratorium Kesehatan', 'Laboratorium', ST_GeomFromText('POINT
  (117.1530 -0.4847)', 4326)),
13 ('Kantin Utama', 'Kuliner', ST_GeomFromText('POINT(117.1545
  -0.4852)', 4326)),
14 ('Kantin Belakang', 'Kuliner', ST_GeomFromText('POINT(117.1548
  -0.4865)', 4326)),
15 ('Gerbang Utama (Juanda)', 'Akses', ST_GeomFromText('POINT(117.1525
  -0.4860)', 4326)),
16 ('Gerbang Samping', 'Akses', ST_GeomFromText('POINT(117.1550
  -0.4850)', 4326)),
17 ('Pos Satpam Depan', 'Keamanan', ST_GeomFromText('POINT(117.1526
  -0.4859)', 4326)),
18 ('Pos Satpam Belakang', 'Keamanan', ST_GeomFromText('POINT(117.1551
  -0.4851)', 4326)),
19 ('Lapangan Basket', 'Olahraga', ST_GeomFromText('POINT(117.1540
  -0.4845)', 4326)),
20 ('Lapangan Futsal', 'Olahraga', ST_GeomFromText('POINT(117.1542
  -0.4846)', 4326)),
21 ('Gedung Puskom', 'Fasilitas', ST_GeomFromText('POINT(117.1527
  -0.4849)', 4326)),
22 ('Auditorium', 'Fasilitas', ST_GeomFromText('POINT(117.1532
  -0.4842)', 4326)),
23 ('Area Parkir Mobil', 'Parkir', ST_GeomFromText('POINT(117.1520
  -0.4855)', 4326)),
24 ('Area Parkir Motor', 'Parkir', ST_GeomFromText('POINT(117.1522
  -0.4862)', 4326)),
25 ('ATM Center', 'Finansial', ST_GeomFromText('POINT(117.1527
  -0.4858)', 4326)),
26 ('Student Center', 'Fasilitas', ST_GeomFromText('POINT(117.1544
  -0.4858)', 4326));

```

4.2 Dataset Jalan Sekitar (10 LineStrings)

```

1 INSERT INTO jalan_samarinda (nama_jalan, geom) VALUES
2 ('Jl. Ir. H. Juanda', ST_GeomFromText('LINESTRING(117.1510 -0.4870,
  117.1560 -0.4840)', 4326)),
3 ('Jl. Kadrie Oening', ST_GeomFromText('LINESTRING(117.1500 -0.4830,
  117.1540 -0.4835)', 4326)),
4 ('Jl. Antasari', ST_GeomFromText('LINESTRING(117.1560 -0.4840,
  117.1580 -0.4820)', 4326)),

```

```

5 ('Jl. Air Hitam', ST_GeomFromText('LINESTRING(117.1540 -0.4835,
    117.1530 -0.4810)', 4326)),
6 ('Jl. Wijaya Kusuma', ST_GeomFromText('LINESTRING(117.1525 -0.4865,
    117.1515 -0.4885)', 4326)),
7 ('Jl. Letjen Suprpto', ST_GeomFromText('LINESTRING(117.1555
    -0.4855, 117.1575 -0.4875)', 4326)),
8 ('Akses Internal Barat', ST_GeomFromText('LINESTRING(117.1526
    -0.4859, 117.1529 -0.4848)', 4326)),
9 ('Akses Internal Timur', ST_GeomFromText('LINESTRING(117.1540
    -0.4845, 117.1545 -0.4852)', 4326)),
10 ('Jl. Gang UMKT 1', ST_GeomFromText('LINESTRING(117.1535 -0.4865,
    117.1545 -0.4875)', 4326)),
11 ('Jl. Gang Perintis', ST_GeomFromText('LINESTRING(117.1520 -0.4840,
    117.1530 -0.4842)', 4326));

```

4.3 Dataset Area/Zona UMKT (Polygon)

```

1 INSERT INTO batas_wilayah (nama_area, geom) VALUES
2 ('Zona Gedung A', ST_GeomFromText('POLYGON((117.1530 -0.4850,
    117.1532 -0.4850, 117.1532 -0.4853, 117.1530 -0.4853, 117.1530
    -0.4850))', 4326)),
3 ('Zona Gedung B', ST_GeomFromText('POLYGON((117.1532 -0.4854,
    117.1534 -0.4854, 117.1534 -0.4857, 117.1532 -0.4857, 117.1532
    -0.4854))', 4326)),
4 ('Zona Gedung C', ST_GeomFromText('POLYGON((117.1534 -0.4857,
    117.1536 -0.4857, 117.1536 -0.4860, 117.1534 -0.4860, 117.1534
    -0.4857))', 4326)),
5 ('Zona Perpustakaan', ST_GeomFromText('POLYGON((117.1527 -0.4849,
    117.1529 -0.4849, 117.1529 -0.4852, 117.1527 -0.4852, 117.1527
    -0.4849))', 4326)),
6 ('Zona Masjid', ST_GeomFromText('POLYGON((117.1536 -0.4853,
    117.1539 -0.4853, 117.1539 -0.4856, 117.1536 -0.4856, 117.1536
    -0.4853))', 4326)),
7 ('Area Lapangan Basket', ST_GeomFromText('POLYGON((117.1539
    -0.4844, 117.1542 -0.4844, 117.1542 -0.4847, 117.1539 -0.4847,
    117.1539 -0.4844))', 4326)),
8 ('Area Parkir Utama', ST_GeomFromText('POLYGON((117.1519 -0.4853,
    117.1523 -0.4853, 117.1523 -0.4858, 117.1519 -0.4858, 117.1519
    -0.4853))', 4326)),
9 ('Area Kantin', ST_GeomFromText('POLYGON((117.1544 -0.4851,
    117.1547 -0.4851, 117.1547 -0.4854, 117.1544 -0.4854, 117.1544
    -0.4851))', 4326)),
10 ('Zona Hijau Depan', ST_GeomFromText('POLYGON((117.1526 -0.4861,
    117.1528 -0.4861, 117.1528 -0.4863, 117.1526 -0.4863, 117.1526
    -0.4861))', 4326)),
11 ('Taman Tengah', ST_GeomFromText('POLYGON((117.1533 -0.4851,
    117.1535 -0.4851, 117.1535 -0.4853, 117.1533 -0.4853, 117.1533
    -0.4851))', 4326)),
12 ('Zona Lab Kes', ST_GeomFromText('POLYGON((117.1529 -0.4846,
    117.1531 -0.4846, 117.1531 -0.4848, 117.1529 -0.4848, 117.1529
    -0.4846))', 4326)),

```

```

13 ('Gedung Puskom Area', ST_GeomFromText('POLYGON((117.1526 -0.4848,
    117.1528 -0.4848, 117.1528 -0.4850, 117.1526 -0.4850, 117.1526
    -0.4848))', 4326)),
14 ('Auditorium Area', ST_GeomFromText('POLYGON((117.1531 -0.4841,
    117.1534 -0.4841, 117.1534 -0.4844, 117.1531 -0.4844, 117.1531
    -0.4841))', 4326)),
15 ('Zona Gerbang Juanda', ST_GeomFromText('POLYGON((117.1524 -0.4859,
    117.1527 -0.4859, 117.1527 -0.4862, 117.1524 -0.4862, 117.1524
    -0.4859))', 4326)),
16 ('Area Parkir Dosen', ST_GeomFromText('POLYGON((117.1538 -0.4860,
    117.1540 -0.4860, 117.1540 -0.4862, 117.1538 -0.4862, 117.1538
    -0.4860))', 4326)),
17 ('Student Center Area', ST_GeomFromText('POLYGON((117.1543 -0.4857,
    117.1546 -0.4857, 117.1546 -0.4860, 117.1543 -0.4860, 117.1543
    -0.4857))', 4326)),
18 ('Lahan Kosong Utara', ST_GeomFromText('POLYGON((117.1520 -0.4835,
    117.1525 -0.4835, 117.1525 -0.4838, 117.1520 -0.4838, 117.1520
    -0.4835))', 4326)),
19 ('Lahan Kosong Timur', ST_GeomFromText('POLYGON((117.1550 -0.4845,
    117.1555 -0.4845, 117.1555 -0.4848, 117.1550 -0.4848, 117.1550
    -0.4845))', 4326)),
20 ('Area Helipad (Simulasi)', ST_GeomFromText('POLYGON((117.1540
    -0.4872, 117.1542 -0.4872, 117.1542 -0.4874, 117.1540 -0.4874,
    117.1540 -0.4872))', 4326)),
21 ('Batas Kampus Terluar', ST_GeomFromText('POLYGON((117.1515
    -0.4840, 117.1555 -0.4840, 117.1555 -0.4880, 117.1515 -0.4880,
    117.1515 -0.4840))', 4326));

```

5 Latihan 2: Analisis Spasial Lokal

5.1 Skenario 1: Pencarian Radius (Nearest Neighbor)

Tampilkan fasilitas UMKT yang berada dalam radius 200 meter dari gerbang depan.

```

1 SELECT nama_tempat
2 FROM fasilitas_umkt
3 WHERE ST_DWithin(geom::geography, ST_GeogFromText('POINT(117.1530
    -0.4855)'), 200);

```

5.2 Skenario 2: Analisis Luas Lahan

Hitung estimasi luas kawasan utama UMKT dalam satuan meter persegi.

```

1 SELECT nama_area, ST_Area(geom::geography) as luas_m2
2 FROM batas_wilayah WHERE nama_area = 'Kawasan Utama UMKT';

```

6 Tugas Praktikum

1. Gunakan *Overpass Turbo* untuk mengambil 5 titik lokasi UMKM/Warung di sepanjang **Jl. Ir. H. Juanda, Samarinda**.

2. Import data tersebut ke tabel baru `umkm_sekitar_umkt`.
3. Buat kueri SQL untuk menampilkan UMKM mana saja yang jaraknya kurang dari 500 meter dari **Gedung E UMKT**.
4. Kumpulkan kueri SQL dan hasil ekspor data dalam format `.geojson`.